

Advanced Data Communication: Principles and Applications of Satellite

PBIT 206 – Advanced Data Communication
Research Essay

Abstract

The essay will be a thorough analysis of satellite data communication, including its historical development and discussion of its technical principles. The work covers orbital mechanics in Low Earth Orbit (LEO) configurations, as well as Medium Earth Orbit (MEO) and Geostationary Orbit (GEO) configurations. The work also covers frequency spectrum management (C, Ku, and Ka) bands. Signal modulation schemes and link budget analysis are presented to put the physical layer performance of satellite systems into perspective. At the centre of this essay will be a discussion of High-Throughput Satellites (HTS) and the technology that surrounds it of spot beam delivery, which will be referred to herein as the Satellite Zooming. The technology supports frequency reuse and directional coverage, and significantly enhances data throughput. A comparative survey of satellite and terrestrial data communication infrastructures looks into the trade-offs in cost, latency, coverage, and resilience. The essay also explores such challenges as atmospheric attenuation, propagation latency, orbital debris, and cybersecurity vulnerabilities. Lastly, the document touches on the future trends emerging i.e. the integration of satellite systems with sixth-generation (6G) wireless networks and the use of artificial intelligence (AI) to control autonomous satellite operations. The general thesis is that satellite communication, especially HTS architecture is inseparable to the present global data infrastructure, and that it will become an increasingly central component of next-generation connectivity paradigms.

1. Introduction: History of Satellite Communication and its contribution to the current global data infrastructure.

The first formal statement of the concept of the use of artificial satellites as relay stations of electromagnetic signals was made by Arthur C. Clarke in 1945 in his landmark article in the *Wireless World*, which proposed the use of the geostationary orbit as a radio relay platform of global scope. The vision envisaged by Clarke, which many at the time dismissed as a science-fiction fantasy, came to fruition within 20 years. On 10 July 1962, the Bell Telephone Laboratories developed Telstar 1 which had been launched by NASA, transmitting the first live transatlantic television pictures and proving the practical viability of satellite communication (Elbert, 2008). This one event led to a historic technological revolution, which has been continuously transforming the world data infrastructure in the next 60 years.

The early days of satellite communications, which dates to around 1960 to 1980, was characterised by large and costly spacecrafts servicing narrowband voice and limited data traffic. In 1964, the International Telecommunications Satellite Consortium (INTELSAT) was formed and created the first global commercial satellite system and by the 1970s was carrying the majority of international telephone traffic (Pratt et al., 2019). These early geostationary satellites used relatively modest capacities in tens of megabits per second with a primary mode of operation in the C band (4-8 GHz).

The decade of 1980s and 1990s saw a dramatic growth occasioned by the liberalisation of telecommunications markets, miniaturisation of electronics and the advent of digital signal processing. The emergence of the internet created an insatiable need of the high-capacity data connections, which was made possible by the Ku-band satellites. VSAT networks were widespread in this period and enabled business and governments to build their own wide-area networks using satellites (Roddy, 2006). In the 1990s, the introduction of the digital video broadcast standards further stimulated the adoption of satellites in broadcasting, as military and intelligence agencies at the same time developed high-resolution data acquisition systems in classified programmes.

The twenty-first century has been characterized by a qualitative change as opposed to a simply a quantitative change. There are three developments which are particularly transformative. The first reason is that the commercialisation of Ka-band frequencies and introduction of High-Throughput Satellite (HTS) architectures enabled individual spacecraft to deliver multiple terabits of aggregate capacity, compared to the gigabits of legacy systems (Bhattacharjee and Singla, 2019). Second, the advent of large constellations of Low Earth Orbit satellites, most notably SpaceX Starlink and Amazon Project Kuiper, has created an entirely new paradigm of broadband delivery with much lower latency. Third, converged non-terrestrial network (NTN) frameworks are emerging, which are blurring the boundaries between space and terrestrial segments.

The satellite communication nowadays forms the basis of a phenomenal number of important activities in the international data infrastructure. Satellite data links are essential to maritime broadband connections on commercial vessels, inflight connectivity on commercial aviation, precision agriculture using satellite images, disaster-response communications, global positioning and timing systems, and the backhauling of cellular traffic in remote regions all depend fundamentally on satellite data links (Maral et al., 2020). The 2023 State of the Satellite Industry Report published by the Satellite Industry Association estimated the global satellite industry revenues at around 281 billion, showing the economic scope of this industry. It is in this context that a technical mastery of the principles of satellite communication is not just an academic requirement; but is a condition to the informed engagement in the design, deployment and regulation of the global connectivity infrastructure.

The structure of this essay is designed to develop in a progressive way out of basic principles to advanced applications and future directions. In section 2, the key technical principles that govern satellite communication have been analyzed, including orbital mechanics, frequency spectrum management, and signal modulation. Section 3 gives a detailed analysis of the High-Throughput Satellites and the spot beam technology. In section 4, a comparative study is made of satellite versus land communication. The following section (5) deals with the main challenges of the industry. Section 6 ends with a discussion of future directions, specifically the integration of 6G and AI-based autonomy.

2. Technical Principles of Satellite Data Communication.

2.1 Orbital Mechanics: LEO, MEO, and GEO.

The propagation delay, coverage footprint, Doppler shift characteristics, and the number of spacecrafts needed to cover the entire globe of a satellite is greatly determined by its orbital altitude. There are three orbit regimes: Low Earth Orbit (LEO), Medium Earth Orbit (MEO) and Geostationary Earth Orbit (GEO), which represent three engineering trade-offs between competing system requirements.

Geostationary Earth Orbit is in an orbital period that is equal to the rotational period of the Earth, namely 23 hours, 56 minutes, and 4 seconds (Wertz and Larson, 2011). This characteristic brings gigantic operation benefits. One GEO satellite is capable of delivering continuous and uninterrupted coverage of about one-third of the earth surface and no tracking is needed at the ground terminal. As such, GEO satellites have traditionally overtaken commercial broadcasting, VSAT enterprise networks, and government communications.

Nevertheless, the enormous height of GEO presents an inherent physical limitation: one-way propagation delay of about 238280 milliseconds, which results in a round trip time of 477560 milliseconds (Pratt et al., 2019). This latency is due to the finite speed of light (around 299,792 km/s) moving along the 35,786 km distance between Earth and satellite and vice versa. In applications where real-time interactivity is needed e.g. online gaming, voice-over-IP, high-frequency financial trading, and some video conferencing applications this delay is a serious constraint. The path loss due to free space at GEO altitudes is about 196 dB at Ku-band frequencies, which requires high-gain antennas and high-power transponders, further increasing both mass and cost of spacecraft.

Medium Earth Orbit falls in altitudes of about 2,000 to 35,786 kilometres. Navigation satellite systems (GPS (USA), GLONASS (Russia), Galileo (EU), and BeiDou (China)) are the most commercially important MEO applications with orbital altitudes of about 19,100 to 23,200 kilometres with orbital periods of approximately 12 hours. O3b (since 2019, SES O3b mPOWER) is also hosted on MEO, and provides high-throughput, low-latency broadband to maritime, aeronautical, and enterprise customers at 8,062 kilometres altitude (Bhattacharjee and Singla, 2019). MEO satellites due to their non-geostationary nature and requiring larger constellations to achieve full coverage have lower free-space path loss than GEO and one-way latencies of roughly 70100 milliseconds, but they require tracking antennas due to their non-geostationary nature.

A dramatic renaissance in the twenty-first century has been brought about by the dramatic decrease in the cost of launching a rocket due to the technology of reusable rockets. LEO satellites have propagation delays of only 1-4 milliseconds in one way, effectively eliminating the latency issue that has afflicted GEO broadband. Nevertheless, only a small arc of sky is covered by a single LEO satellite with an altitude of 550 kilometres; this satellite passes overhead only about 57 minutes, requiring constellations of hundreds or thousands of satellites to provide a continuous coverage of the earth (Del Portillo et al., 2019). By early 2024, over 5,000 satellites in the Ku and Ka bands had been deployed by SpaceX Starlink, with the majority at 550 km altitude. The Project Kuiper of Amazon is scheduled to have 3,236 satellites. These megaconstellations are a paradigm shift of the conventional model of large expensive spacecraft to many smaller, cheaper, mass-produced spacecraft.

The orbital elements that characterize the path of a satellite are the semi-major axis, the eccentricity, the inclination, the right ascension of the ascending node, the argument of perigee and the true anomaly (Wertz and Larson, 2011). In the case of geostationary satellites, both the eccentricity and

inclination are zero (circular orbit) and zero degrees respectively. Practically, the gravitational perturbation of the Moon, Sun, and the oblateness of the Earth and its orbit requires north-south and east-west station keeping manoeuvres using onboard propulsion, which consumes about 45 m/s of ΔV /annum and limits the operational life of the GEO satellites to a typical of 1518 years. LEO satellites in inclined orbit experience atmospheric drag at lower altitudes, necessitating periodic reboost and imposing a lower limit on operational lifetime, although at altitudes above a few kilometers, atmospheric drag becomes negligible to the operational lifetime of a decade or longer.

2.2 Frequency Spectrum Management: C, Ku and Ka Bands.

The electromagnetic spectrum that is accessible to satellite communication is a limited, internationally regulated resource controlled by the International Telecommunication Union (ITU) through its Radio Regulations. The propagation characteristics, size requirements of antennas, and vulnerability to atmospheric impairments vary across different frequency bands, thus the choice of band is a critical design consideration in satellite system architecture.

The C band (4 -8 GHz, with downlink to the satellite at 3.7-4.2 GHz and uplink at 5.925-6.425 GHz) was the workhorse of early commercial satellite communication and remains very widely used in broadcasting and trunk telephony. The main benefit of C band is that it is relatively immune to rain fade: at these frequencies, precipitation only causes an attenuation of 0.01100 dB/km, which can be easily absorbed by atmospheric factors (Ippolito, 2017). Nevertheless, its relatively long wavelength (compared to other bands) necessitates the use of large dish antennas (usually 1.8-3.7 metres to receive) which has led to its sharing with other bands such as terrestrial microwave links, creating coordination complexity in densely populated regions. The 2020 C-band transition of the United States, which cleared 280 MHz of C-band spectrum to 5G terrestrial use, is an example of the mounting pressure on this frequency resource.

The Ku band (1218 GHz, with satellite broadcast downlinks at 10.712.75 GHz and uplinks at 1414.5 GHz) offers a higher gain with smaller antennas - 6090 cm dishes are the norm with direct-to-home (DTH) television reception - and consumer-grade satellite services at affordable equipment prices (Roddy, 2006). Ku band also has an advantage of less terrestrial interference than C band. The trade-off is a higher susceptibility to rain attenuation: at 12 GHz, specific attenuation under moderate rain (10 mm/h) reaches about 0.510 dB/km, and a very large amount at higher frequencies. The most commonly used frequency is ku band used in broadcast satellite services, maritime and aeronautical VSAT and a large part of LEO broadband constellations.

The frequency of choice in contemporary High-Throughput Satellites has become ka band (26.5 -40 GHz, with downlinks at 18.3 -20.2 GHz and uplinks at 27.5 -30.0 GHz) due to the large amounts of available bandwidth and the possibility of forming small, high-gain spot beams with physically small satellite antennas (Evans et al., 2005). Ka band offers significantly greater spectral bandwidth than Ku band - typically 500 MHz as compared to 250 MHz per transponder - and can provide higher total data rates. But Ka band is much more prone to rain fade: the specific attenuation at 30 GHz is 5-10 dB/km in heavy rain, and it needs effective fade mitigation practices. These consist of Adaptive Coding and Modulation (ACM) which dynamically adjusts the coding rate and modulation order based on real-time estimations of the link quality, and site diversity, where two spatially separated ground terminals are used to combat localised rainfall cells.

In addition to C, Ku and Ka bands, emerging uses are increasingly exploiting V band (4075 GHz), Q band (3350 GHz) and even W band (75110 GHz) in feeder links and inter-satellite links (ISLs). The laser ISLs in SpaceX Starlink, work in optical spectrum providing inter-satellite communication without radio frequency interference. These optical ISLs also allow routing of traffic via the constellation, minimising the use of ground stations and potentially enabling lower-latency transoceanic routing than with terrestrial fibre cables (Del Portillo et al., 2019).

2.3 Signal Modulation and Analysis of Link Budget.

Current satellite systems use advanced digital modulation algorithms to achieve maximum spectral efficiency and still have reasonable bit error rates over very dynamically changing link conditions. The modulation type has a direct impact on the throughput, robustness, and the signal-noise ratio (SNR) required.

Phase-shift keying (PSK) - and especially the types of Binary PSK (BPSK) and Quadrature PSK (QPSK) - were the foundations of early digital satellite modulation. QPSK uses two bits per symbol, by adjusting the carrier phase among four states (0, 90, 180, 270), to offer a spectral efficiency of about 2 bits/Hz at a practical implementation loss. It has good noise immunity with a required E_b/N_0 of about 4.4 dB to achieve a 10^{-6} bit error rate (BER) with ideal coding (Sklar, 2001). Greater numbers of bits per symbol are encoded at the expense of reduced noise margin using higher SNR to achieve acceptable BER. The current state-of-the-art broadcast and broadband satellite modulation is the DVB-S2X standard which supports up to 32APSK (32-state amplitude and phase-shift keying) modulation and uses the Low-Density Parity-Check (LDPC) codes in combination with the Bose-Chaudhuri-Hocquenghem (BCH) codes achieving within 0.7 dB of the theoretical Shannon limit (Morello and Mignone, 2006).

The accounting system that is used to ensure that a satellite system will be able to provide the necessary data rate with the necessary availability given a given set of system parameters is known as the link budget analysis. The basic relation equation is between the received signal power and the transmitted power based on the gains and losses. The carrier-to-noise density ratio (C/N_0) received can be defined as:

$$C/N_0 \text{ [dBHz]} = \text{EIRP [dBW]} - L_{fs} \text{ [dB]} - L_a \text{ [dB]} + G/T \text{ [dB/K]} - k \text{ [dBW/Hz/K]}$$

where EIRP is the effective isotropically radiated power of the transmitter, L_{fs} is the free-space path loss (approximately $92.4 + 20 \cdot \log_{10}(f \text{ GHz}) + 20 \cdot \log_{10}(d \text{ km})$ dB), L_a is atmospheric attenuation, G/T is the receive figure of merit (antenna gain over system noise temperature), and k is Boltzmann constant (-228.6 dBW/Hz/K) (Pratt et al., 2019). The complete link budget should include uplink and downlink margins separately including transponder noise, inter-satellite interference and pointing loss. Narrow spot beam High-Throughput Satellites offer much higher EIRP and G/T compared to wide-beam satellites, and therefore much higher spectral efficiency and lower complexity of earth station equipment.

Adaptive Coding and Modulation (ACM) is an important enabling technology to achieve effective use of satellite capacity in the presence of variable channel conditions. The ground terminal of an ACM system

continuously measures the received SNR, and reports it to the hub, which then adjusts the modulation and coding rate (ModCod) to maximise throughput given the available link margin. With a clear sky, it is possible to use a high ModCod like 32APSK 9/10 that provides almost 5 bits/symbol. . During rain fade, the system gracefully degrades to a more robust ModCod such as QPSK 1/4, trading throughput for link reliability. DVB-S2X defines 116 distinct ModCods, providing fine-grained adaptation across a wide range of link conditions (ETSI EN 302 307-2, 2015).

3. Satellite Zooming and Precision: High-Throughput Satellites and Spot Beam Technology

3.1 Architecture of High-Throughput Satellites

The term 'Satellite Zooming' as used in this essay refers to the capability of HTS systems to concentrate radio frequency energy into narrow, geographically targeted beams — analogous to the zoom function of an optical lens. By focusing transmitted power into a smaller area, these systems achieve dramatically higher power flux density at the receive terminal, enabling greater spectral efficiency and higher user data rates. This section provides a technical analysis of how HTS achieve this capability and the engineering principles that underpin it.

A conventional wide-beam GEO satellite radiates its power across a continental or hemispheric coverage zone. The effective isotropically radiated power (EIRP) is therefore distributed over a

very large surface area, and any given ground terminal receives only a tiny fraction of the total transmitted power. A High-Throughput Satellite, by contrast, uses a large phased array or shaped reflector antenna system to generate tens to hundreds of narrow spot beams, each covering a relatively small geographic area — typically 200–600 kilometres in diameter, depending on the orbital altitude and antenna aperture (Evans et al., 2005).

The key architectural innovation enabling HTS is aggressive frequency reuse. In a conventional satellite, the entire allocated bandwidth is used once across the coverage area. In an HTS system, the available spectrum is divided into a set of frequency/polarisation combinations — commonly four colours using two frequency segments and two orthogonal polarisations (LHCP and RHCP) — and each colour is reused in spatially separated beams that have sufficient angular separation to limit inter-beam interference to acceptable levels (Fenech et al., 2016). This four-colour frequency reuse scheme allows the satellite to offer four times the capacity of a single-beam system using the same total spectrum allocation. More advanced systems use three-colour or even full-frequency-reuse architectures, where adjacent beams share spectrum and the interference is managed through sophisticated beam-shaping and pre-coding techniques.

To illustrate the capacity gains achievable through spot beam frequency reuse, consider the ViaSat-3 spacecraft, launched in 2023. This satellite uses approximately 1,000 spot beams across three orbital positions to deliver aggregate throughput exceeding 1 Tbps, compared with the 100 Gbps aggregate capacity of legacy wide-beam GEO satellites of similar mass (Viasat Inc., 2023). This tenfold capacity improvement is achieved primarily through frequency reuse and spot beam concentration of EIRP, without requiring a proportional increase in spacecraft transmit power. The O3b mPOWER constellation achieves similar benefits at MEO altitude using steerable spot beams with electronically controlled phased array antennas, enabling dynamic beam repositioning to follow traffic demand in real time.

The antenna systems aboard HTS represent one of the most technically demanding aspects of spacecraft engineering. Generating 100+ simultaneous spot beams requires either a complex multi-feed reflector system — where many small feed horns illuminate a single large parabolic reflector — or an active phased array antenna (APAA) where thousands of individual radiating elements are independently controlled in amplitude and phase. Multi-feed reflector systems are simpler and more power-efficient but have fixed beam positions. Active phased arrays enable dynamic beam

steering and shaping but consume significantly more direct current power and generate substantial waste heat requiring active thermal management (Vidal et al., 2017).

3.2 Frequency Reuse and Inter-Beam Interference Management

The fundamental challenge of spot beam frequency reuse is managing interference between adjacent beams sharing the same spectrum. The isolation between beams depends on the antenna sidelobe pattern: an ideal antenna would have infinitely sharp roll-off beyond the beam edge, but practical antennas exhibit gradual sidelobe degradation. The carrier-to-interference ratio (C/I) between a desired beam and an interfering co-channel beam determines the maximum achievable ModCod and hence the capacity of each beam.

For a four-colour frequency reuse pattern, adjacent beams use different frequency/polarisation combinations, and the nearest co-channel interferers are separated by approximately 2–3 beam diameters. This spatial separation, combined with the antenna pattern roll-off, typically yields C/I ratios of 12–20 dB, which is sufficient to support efficient high-order modulation (Fenech et al., 2016). Advanced interference mitigation techniques, including precoding and multi-user MIMO (MU-MIMO) applied across the beam set, can substantially increase this ratio by exploiting the spatial degrees of freedom available in a multi-beam antenna system. Precoding, adapted from

terrestrial cellular communication theory, applies a linear or non-linear transformation to the transmitted signal vector such that inter-beam interference is cancelled at the receive terminal, enabling full frequency reuse with acceptable C/I.

The design of the beam layout — the tessellation of spot beams across the coverage area — is a complex optimisation problem balancing uniform service quality, traffic demand distribution, available spectrum, and co-channel interference constraints. In practice, traffic demand is highly non-uniform: dense urban areas may require five to ten times the capacity of rural regions. Modern HTS systems address this through flexible payload architectures that allow reallocation of power and bandwidth between beams via software command from the ground, enabling dynamic adaptation to shifting demand patterns over the satellite's operational lifetime (Vidal et al., 2017).

3.3 High-Resolution Data Acquisition via Satellite

Beyond communication relay, satellite systems serve as critical platforms for high-resolution data acquisition — collecting imagery, remote sensing data, and scientific measurements that cannot be obtained from the ground. The 'Satellite Zooming' concept in this context refers to the use of high-resolution optical and radar sensors to acquire detailed information about specific geographic targets, analogous to zooming in with a camera.

Commercial Earth observation satellites have achieved remarkable resolution advances in recent years. Maxar Technologies' WorldView-3, operating at 617 km altitude, provides panchromatic resolution of 31 centimetres — sufficient to identify individual vehicles in a car park and distinguish between car models (DigitalGlobe, 2016). Synthetic Aperture Radar (SAR) satellites such as the European Space Agency's Sentinel-1 and commercial ICEYE constellation provide all-weather, day-and-night imaging with 1–25 metre resolution, essential for maritime surveillance, flood mapping, and infrastructure monitoring (Moreira et al., 2013).

The data volumes generated by these systems are immense. A single WorldView-3 image covering a 13.1×13.1 km area in 8-band multispectral mode generates approximately 2.5 GB of raw data before compression. Downlinking this data to ground stations requires high-data-rate optical or X-band radio links, with some systems achieving downlink rates of several gigabits per second. The European Space Agency's EDRS (European Data Relay System) uses optical inter-satellite links to relay data from polar-orbiting satellites to geostationary relay spacecraft, which then downlink to ground stations, dramatically increasing the data return capacity of LEO Earth observation missions (ESA, 2019).

The integration of AI and machine learning directly on the satellite — known as onboard processing or edge computing — is transforming high-resolution data acquisition. Rather than downlinking all raw imagery, a satellite equipped with a neural processing unit can perform onboard change detection, object recognition, and data compression, transmitting only the high-value extracted information. DARPA's Blackjack programme and several commercial missions have demonstrated this capability, reducing the required downlink bandwidth by orders of magnitude while increasing the actionability of delivered information (Roper et al., 2020).

4. Comparative Analysis: Satellite vs. Terrestrial Data Communication

4.1 Coverage and Accessibility

The most fundamental advantage of satellite communication over terrestrial alternatives is coverage ubiquity. Terrestrial networks — fibre optic cables, microwave links, and cellular base stations — require physical infrastructure deployment, which is economically viable only where population density justifies the capital expenditure. Approximately 40% of the world's population remains unserved or underserved by terrestrial broadband infrastructure, primarily in rural sub-Saharan Africa, Southeast Asia, and remote regions of the Americas and Oceania (ITU, 2023). Satellite systems can serve these populations without any in-country ground infrastructure beyond a user terminal and power source, making them uniquely suited to bridging the digital divide.

The economics of coverage extension differ fundamentally between satellite and terrestrial approaches. Deploying fibre optic cable to a rural community requires trenching, installing conduit, and splicing fibre — costs that scale linearly or super-linearly with distance. Connecting that same community via satellite requires only a ground terminal (currently approximately \$299–\$599 for Starlink consumer equipment) and a subscription, with no marginal infrastructure cost for the network operator regardless of the terminal's location (SpaceX, 2024). This asymmetric cost structure strongly favours satellite for sparse population distributions and remote or difficult terrain, including maritime and aeronautical applications.

4.2 Throughput and Latency

Terrestrial fibre optic networks currently offer the highest available data throughput at the lowest latency. Single-mode silica fibre, operating at near-infrared wavelengths around 1,550 nm using Dense Wavelength Division Multiplexing (DWDM), can carry multiple terabits per second over a single fibre pair, with propagation delay determined by the speed of light in glass — approximately 200,000 km/s (compared with 299,792 km/s in free space). A trans-Pacific fibre cable covering approximately 14,000 km introduces approximately 70 ms one-way latency, which is competitive with MEO satellite links and far superior to GEO.

GEO satellite broadband systems, with their inherent 238–280 ms one-way delay, are fundamentally unsuitable for certain latency-sensitive applications. The TCP protocol's congestion control mechanisms — designed for the sub-100 ms round-trip times of terrestrial networks — perform poorly over GEO links, requiring Performance Enhancing Proxies (PEPs) that split the TCP connection at the satellite terminal to improve throughput efficiency (Caini et al., 2012). Voice-over-IP over GEO satellite is technically feasible but results in noticeable conversational delay that degrades user experience. Interactive gaming and financial transaction systems requiring sub-10 ms latency are incompatible with GEO links.

LEO broadband constellations have substantially narrowed the latency gap with terrestrial networks. Starlink reports median latency of 20–40 ms in its standard service tier, comparable to a moderate-quality terrestrial DSL or cable connection (Bhattacharjee & Singla, 2019). However, peak throughput per user in LEO systems remains constrained by spectrum availability and satellite capacity allocation, with typical Starlink users experiencing 50–200 Mbps download speeds compared with the gigabit speeds available via fibre. As LEO constellation densities increase and onboard processing improves, this throughput gap is expected to narrow progressively.

4.3 Resilience and Disaster Recovery

Satellite communication offers inherent resilience advantages that terrestrial networks cannot match for certain scenarios. Terrestrial infrastructure — cables, conduits, and base stations — is vulnerable to physical disruption by natural disasters (earthquakes, floods, hurricanes), human conflict, and accidental damage. A major earthquake can sever hundreds of underground fibre runs simultaneously, creating communication blackouts in exactly the regions where communication is most desperately needed for emergency response.

Satellite systems are physically isolated from terrestrial disasters and can maintain continuous service over affected regions. The value of this resilience was dramatically demonstrated following the 2011 Tōhoku earthquake and tsunami in Japan, where satellite terminals provided the only operational communication links in many coastal communities for days after the disaster (Kawamoto et al., 2019). Similarly, during Hurricane Maria's devastation of Puerto Rico's terrestrial infrastructure in 2017, satellite links provided essential connectivity for emergency response coordination. The 2022 Russian invasion of Ukraine prompted SpaceX to deliver Starlink terminals to Ukrainian military and civilian users, providing resilient broadband connectivity despite extensive terrestrial infrastructure damage, illustrating satellite communication's strategic importance in conflict scenarios.

4.4 Cost and Economics

The economics of satellite versus terrestrial communication depend heavily on traffic volumes, geographic distribution, and service requirements. For high-density urban environments, terrestrial fibre offers dramatically lower cost per bit — the one-time infrastructure investment is amortised over millions of users. The cost per megabyte of data transit on terrestrial backbone networks has fallen below \$0.001 in many markets. Satellite capacity, while cheaper than ever, remains significantly more expensive per bit than terrestrial fibre, with typical HTS Ka-band capacity pricing of \$100–\$500 per MHz per month (depending on region and traffic type) (Maral et al., 2020).

The satellite economics are improving rapidly. The combination of HTS frequency reuse achieving higher capacity per spacecraft, lower launch costs from reusable rockets, and economies of scale from mass-produced LEO satellites has driven satellite bandwidth costs down by approximately 90% over the period 2010–2023. The cost per gigabyte of Starlink data, at the \$120/month consumer price point with typical usage of 200–500 GB per month, is broadly comparable with mobile data in many developed countries. For applications requiring global mobility — maritime, aeronautical, or expeditionary — satellite frequently represents not merely the most affordable option but the only technically feasible option.

5. Challenges in Satellite Data Communication

5.1 Atmospheric Attenuation

Atmospheric attenuation — the absorption and scattering of radio waves by atmospheric constituents — represents one of the fundamental physical challenges in satellite link design, particularly at higher frequencies. The atmosphere's effect on satellite signals comprises several components: gaseous absorption by oxygen and water vapour, rain attenuation, cloud and fog attenuation, tropospheric scintillation, and ionospheric effects.

Gaseous absorption in the troposphere primarily results from oxygen (O₂) and water vapour (H₂O) molecules, which exhibit absorption resonances at specific frequencies. Oxygen has a major absorption band around 60 GHz (critical for millimetre-wave applications) and a smaller resonance at 118.75 GHz. Water vapour has resonances at 22.235 GHz and 183.31 GHz. At Ka-band frequencies (26.5–40 GHz), gaseous absorption is relatively modest at approximately 0.1–0.3 dB per kilometre of zenith path in humid conditions, but accumulates to 0.5–1.5 dB over the full tropospheric path (Ippolito, 2017).

Rain attenuation is the dominant impairment at Ku and Ka bands in many regions. Rain drops absorb and scatter electromagnetic energy, with the attenuation coefficient scaling strongly with rain rate and frequency. The ITU-R P.838 model specifies the specific attenuation (dB/km) as a

function of rain rate (mm/h) and frequency. At 30 GHz, specific attenuation in moderate rain (20 mm/h) is approximately 3 dB/km, rising to 8 dB/km in heavy rain (50 mm/h). Combined with an effective rain height of 3–5 km, this implies link margins of 5–25 dB may be required to maintain service at the required availability in tropical regions — equivalent to several times the clear-sky signal level (Maral et al., 2020).

Fade mitigation techniques employed to counteract rain attenuation include: uplink power control (ULPC), which increases the transmitted power during rain events; ACM, which reduces the modulation order and coding rate to maintain a more robust link; site diversity, which exploits the spatial decorrelation of rainfall by routing traffic through an alternate ground terminal located outside the rain cell; and frequency diversity, which temporarily uses a lower-frequency (less rain-susceptible) link when the primary Ka-band link experiences severe fade (Fenech et al., 2016). These techniques collectively allow Ka-band systems to achieve 99.5–99.9% link availability in most climatic regions at the cost of additional system complexity and headroom.

5.2 Propagation Latency

Propagation latency in GEO satellite systems (477–600 ms round-trip) imposes fundamental constraints on applications and protocols that are difficult to overcome through engineering

innovation. Unlike terrestrial latency, which results from the finite processing speed of routers and the physical distance between nodes, satellite latency is an inescapable consequence of the speed of light and the orbital altitude. No amount of onboard processing improvement or ground system optimisation can reduce the electromagnetic transit time.

The impact on TCP performance is particularly significant. TCP's slow-start algorithm increases the congestion window gradually, probing for available bandwidth. Over a GEO link with 600 ms round-trip time, filling a 100 Mbps pipe requires a receive window of approximately 7.5 megabytes — far exceeding the default TCP receive window of 64 kilobytes. Without window scaling (RFC 7323) or Performance Enhancing Proxies, throughput over GEO satellite links is severely constrained by the bandwidth-delay product (Caini et al., 2012). QUIC, the transport protocol underlying HTTP/3, offers potential advantages over GEO links through its connection establishment efficiency and multiplexing capabilities, but the fundamental round-trip time penalty remains.

LEO constellations largely resolve the latency problem for broadband users, but introduce new challenges related to satellite handover management. As a LEO satellite passes overhead in 5–7 minutes, the user terminal must hand over active connections to the next satellite in the

constellation. Seamless handover without packet loss or session disruption requires precise timing, predictive beam management, and coordination between adjacent satellites — a substantial systems engineering challenge for constellations of thousands of spacecraft (Del Portillo et al., 2019).

5.3 Orbital Debris

The proliferation of satellites, particularly the emergence of megaconstellations, is creating an increasingly congested orbital environment that poses risks both to individual spacecraft and to the long-term sustainability of orbital zones. The European Space Agency's 2023 Space Environment Report estimated over 30,000 trackable objects larger than 10 cm in Earth orbit, with potentially millions of smaller but equally lethal fragments from previous collision and fragmentation events (ESA, 2023).

The catastrophic potential of uncontrolled debris proliferation was analysed by Kessler and Cour-Palais (1978) in what has become known as the Kessler Syndrome: a scenario where the density of objects in LEO becomes sufficient that collisions generate additional debris at a rate exceeding atmospheric deorbiting, leading to a self-sustaining cascade that renders certain orbital shells unusable for decades or centuries. While the Kessler threshold has not yet been reached, several

events — including China's 2007 anti-satellite missile test and the 2009 Iridium-Cosmos collision — have substantially increased the LEO debris population (Liou & Johnson, 2009).

The addition of thousands of Starlink, Kuiper, and OneWeb satellites introduces both risk and, somewhat paradoxically, mitigation measures. SpaceX's Starlink satellites are equipped with autonomous collision avoidance systems that execute manoeuvres based on conjunction warnings from the US Space Surveillance Network. Starlink satellites are also designed to deorbit within five years of end-of-life, well within the 25-year guideline established by the Inter-Agency Space Debris Coordination Committee (IADC). However, the scale of megaconstellations means that even a small failure rate of automatic deorbit systems could leave thousands of dead satellites in orbit (Boley & Byers, 2021).

5.4 Cybersecurity Vulnerabilities

Satellite communication systems present a distinctive cybersecurity attack surface that encompasses the space segment, ground segment, and user terminal, as well as the radio frequency (RF) links connecting them. The geographic broadcast nature of satellite signals — which by definition cover large areas and cannot be physically confined to authorised receivers — creates inherent interception vulnerabilities that differ fundamentally from point-to-point terrestrial links.

RF-layer attacks include signal jamming, where a sufficiently powerful interfering signal on the same frequency overwhelms legitimate satellite signals at the receiver; spoofing, where a counterfeit signal mimicking the satellite is transmitted to deceive the receiver into calculating a false position or accepting false data; and signal interception, where unencrypted downlinks are captured by an adversary's ground station (Humphreys et al., 2008). The vulnerability of GPS to spoofing has been demonstrated in multiple operational contexts, including maritime navigation incidents in the Black Sea region and drone interdiction operations.

The 2022 Viasat KA-SAT cyberattack, which occurred concurrent with Russia's invasion of Ukraine, demonstrated the devastating potential of cyber operations against satellite ground systems. The attack exploited a misconfiguration in the SATCOM terminal management system to overwrite the firmware of tens of thousands of satellite modems across Europe, rendering them permanently inoperable (Guerrero-Saade & Guerrero-Saade, 2022). The attack disrupted communications for Ukrainian military units and caused collateral damage to wind farm control systems in Germany. This incident underscored the critical importance of robust authentication, encrypted management protocols, and supply-chain security in satellite ground system software.

Best-practice cybersecurity for satellite systems now encompasses: end-to-end encryption of all user traffic using AES-256 or equivalent, with key management conducted through secure ground infrastructure; strong mutual authentication between user terminals and network operations centres; software-defined radio designs that can receive over-the-air firmware updates through cryptographically signed update packages; intrusion detection and anomaly monitoring applied to telemetry, tracking, and command (TT&C) links; and redundant, geographically dispersed ground station infrastructure to eliminate single points of failure (Falco et al., 2018).

6. Future Trends: 6G Integration and AI-Driven Autonomous Operations

6.1 Integration of Satellite Networks with 6G

The sixth generation of mobile communication standards (6G), expected to be commercially deployed between 2030 and 2035, is being conceptualised from its earliest design phases as a heterogeneous network that natively integrates terrestrial and non-terrestrial components. This represents a fundamental departure from the 4G and early 5G approach, where satellite connectivity was treated as a separate, supplementary layer rather than an integrated component of the core network architecture.

The 3rd Generation Partnership Project (3GPP) introduced Non-Terrestrial Network (NTN) specifications into the 5G standard with Release 17 (completed 2022), enabling 5G New Radio

(NR) protocols to operate over GEO and LEO satellite links. This groundwork is being extended in Release 18 and beyond to support transparent integration of LEO broadband satellites as radio access network nodes, enabling seamless handover between terrestrial base stations and satellite links without service disruption (Lin et al., 2021). For 6G, the ITU-R IMT-2030 framework envisions coverage extension as a primary use case, explicitly incorporating satellite networks as integral components rather than supplementary overlays.

The technical challenges of 6G-satellite integration are substantial. Doppler shifts induced by LEO satellite motion reach several kilohertz at 5G NR frequencies, requiring Doppler pre-compensation at the satellite or user terminal to prevent inter-carrier interference in OFDM systems. Timing advance management — the mechanism that synchronises multiple users' transmissions to arrive at the satellite without overlapping — must account for propagation delays of 1–300 ms depending on orbital altitude, far exceeding the terrestrial timing advance range (Lin et al., 2021). The integration of satellite and terrestrial core networks requires unified authentication, mobility management, and quality-of-service frameworks that can transparently handle the handover between radio access technologies.

The anticipated use cases for integrated 6G satellite networks include: seamless global coverage for massive IoT deployments in agriculture, logistics, and environmental monitoring; ultra-reliable low-latency connectivity for autonomous vehicle platoons and drone swarm coordination in areas beyond terrestrial cellular coverage; global telepresence and extended reality applications requiring multi-gigabit downlinks and sub-100 ms round-trip latency; and integrated sensing and communication, where satellite radar functionality is co-located with broadband communication payloads on the same spacecraft (Saad et al., 2020). Terahertz communication, a candidate technology for 6G's highest data rate applications, may find application in satellite inter-link communications at close-range inter-satellite distances where atmospheric absorption is not a limiting factor.

Software-Defined Networking (SDN) and Network Function Virtualisation (NFV) are critical enabling technologies for 6G satellite integration. By abstracting network control from the underlying hardware and implementing network functions as software on commodity processors, SDN/NFV enables dynamic reconfiguration of satellite network capacity, routing, and quality-of-service policies in response to changing traffic demands and link conditions. This flexibility is

essential in a heterogeneous 6G environment where traffic must be intelligently routed between satellite and terrestrial segments based on real-time network state (Ferrus et al., 2016).

6.2 AI-Driven Satellite Autonomous Operations

Artificial intelligence and machine learning are transforming satellite operations across the entire spacecraft lifecycle, from design and manufacturing through launch, on-orbit operations, and end-of-life disposal. The increasing scale and complexity of satellite constellations — with thousands of spacecraft requiring continuous monitoring, collision avoidance, and capacity management — makes traditional manual operations economically and technically infeasible, creating a powerful imperative for AI-driven autonomy.

In the domain of capacity and resource management, machine learning algorithms are being deployed to predict traffic demand patterns with hourly to seasonal granularity, enabling proactive allocation of satellite capacity to high-demand regions before congestion occurs. Reinforcement learning approaches, trained on historical traffic data and real-time network telemetry, can optimise beam power allocation, frequency assignment, and gateway routing simultaneously across hundreds of spot beams, a problem that is computationally intractable using conventional optimisation methods due to its combinatorial complexity (Roper et al., 2020). Initial deployments

by SES, Intelsat, and several LEO operators have demonstrated 10–25% efficiency improvements in capacity utilisation compared with static allocation schemes.

Autonomous collision avoidance is perhaps the most safety-critical application of AI in satellite operations. The US Space Surveillance Network tracks approximately 27,000 objects and issues thousands of conjunction warnings daily to satellite operators. Evaluating each warning, computing avoidance manoeuvre options, and executing the optimal manoeuvre within the available decision window requires computational capabilities and decision speed that exceed human operator capacity at constellation scale. SpaceX's Starlink satellites are equipped with autonomous collision avoidance algorithms that execute manoeuvres without ground intervention when conjunction probability exceeds 1 in 10,000, processing orbital mechanics calculations and manoeuvrability constraints in real time (Boley & Byers, 2021).

Predictive maintenance and anomaly detection represent another high-value AI application in satellite operations. Satellite subsystems — solar panels, batteries, attitude control actuators, thermal control systems — generate continuous telemetry streams that indicate their health state. Machine learning models trained on historical telemetry from large satellite fleets can identify subtle degradation patterns that precede component failure by days or weeks, enabling pre-emptive

operational mode changes that extend spacecraft life. Given that a GEO satellite represents a capital investment of \$200–\$400 million, extending operational life by even six months through AI-enhanced maintenance can deliver significant economic value (Falco et al., 2018).

The concept of in-orbit servicing — where a robotic servicer spacecraft performs refuelling, component replacement, or orbit adjustment on a client satellite — is increasingly feasible and is being pursued commercially by Northrop Grumman's Mission Extension Vehicle and satellite servicing ventures. AI-driven autonomous rendezvous, docking, and manipulation systems are critical enablers of this capability, as the communication latency to a geostationary satellite (600 ms round-trip) makes teleoperations from Earth impractical for precision manoeuvres. Machine learning-based computer vision systems that can autonomously guide a servicer spacecraft to a tumbling, non-cooperative client satellite represent one of the most ambitious current applications of AI in the space domain (Roper et al., 2020).

Looking further ahead, the concept of cognitive satellite networks envisions spacecraft that autonomously adapt their communication parameters, routing strategies, and resource allocation in response to the radio environment, network state, and user demand — without continuous ground intervention. Such systems would leverage federated machine learning, where models are

trained across multiple satellites and ground stations without centralising sensitive user data, in combination with edge inference capabilities onboard the satellite. The computational power available on modern satellite processors — increasingly based on radiation-hardened versions of commercial GPU architectures — makes onboard inference of sophisticated neural network models increasingly practical (Saad et al., 2020).

7. Conclusion

This essay has provided a comprehensive examination of satellite data communication, addressing its historical evolution, technical foundations, advanced capabilities, comparative position relative to terrestrial networks, operational challenges, and future trajectory. The analysis demonstrates that satellite communication has undergone a remarkable transformation from the narrowband voice relay systems of the 1960s to the multi-terabit High-Throughput Satellites and low-latency LEO broadband constellations of the present decade.

The technical principles underlying satellite communication — orbital mechanics across LEO, MEO, and GEO, frequency spectrum management in C, Ku, and Ka bands, and sophisticated digital modulation with adaptive coding — form an interconnected system in which choices in one domain constrain and inform choices in others. The GEO orbital slot provides coverage continuity

but imposes an inescapable latency penalty; LEO constellations eliminate the latency problem but require thousands of spacecraft and complex handover management. Ka-band frequencies enable narrow spot beams and high throughput but demand robust rain fade mitigation. These engineering trade-offs are not deficiencies to be overcome but fundamental parameters to be optimised within the constraints of physics.

The concept of Satellite Zooming — High-Throughput Satellite spot beam technology and frequency reuse — represents the most significant architectural innovation in commercial satellite communication of the past two decades. By concentrating RF energy into narrow geographic beams and reusing the available spectrum across spatially separated beams, HTS systems achieve capacity improvements of one to two orders of magnitude compared with conventional wide-beam architectures. The forthcoming generation of flexible digital payload satellites, capable of dynamically repositioning, reshaping, and reallocating power to individual beams via software command, will further extend this advantage and enable satellite systems to efficiently serve the highly non-uniform and temporally variable traffic demands of the twenty-first century.

The comparative analysis reveals that satellite and terrestrial networks are increasingly complementary rather than competitive. Terrestrial fibre and cellular networks serve dense

populations with the highest throughput at the lowest cost per bit; satellite networks provide the global coverage layer that terrestrial infrastructure cannot economically replicate. The integration of these complementary technologies in 6G architectures — through 3GPP NTN specifications and SDN/NFV-enabled multi-path routing — represents the logical culmination of this complementarity into a unified global communication infrastructure.

The challenges facing satellite communication — atmospheric attenuation, propagation latency, orbital debris, and cybersecurity — are real and in some cases fundamental. However, the engineering community has demonstrated consistent ingenuity in developing mitigation strategies: ACM and site diversity for atmospheric impairments; TCP acceleration and QUIC for latency; responsible constellation design and debris mitigation standards for the space environment; and comprehensive layered security frameworks for cybersecurity. Regulatory frameworks must evolve in parallel with technology to ensure the orbital environment remains accessible for future generations.

The integration of artificial intelligence throughout satellite systems — from onboard autonomous operations and predictive maintenance to network-wide resource optimisation and cognitive radio adaptation — is accelerating the maturation of these technologies and unlocking capabilities that

were inconceivable with conventional deterministic control systems. The symbiosis between AI and satellite technology will define the character of space-based communication in the coming decades, potentially enabling constellations of thousands of spacecraft to operate with the operational efficiency and adaptability of a single well-managed system.

In conclusion, satellite data communication has progressed from a supplementary telecommunications tool to an indispensable pillar of global digital infrastructure. Its continuing evolution — driven by HTS architectures, LEO megaconstellations, 6G integration, and AI autonomy — positions satellite communication as a central enabler of the connected, data-driven world that the twenty-first century demands. A thorough understanding of the principles, capabilities, and challenges of satellite data communication is therefore not merely academically rewarding but practically essential for any engineer or policymaker engaged with global connectivity.

References

- Bhattacharjee, D., & Singla, A. (2019). Network topology design at 27,000 km/hour. Proceedings of the 15th International Conference on Emerging Networking Experiments and Technologies (CoNEXT 2019), 341–354. <https://doi.org/10.1145/3359989.3365407>
- Boley, A. C., & Byers, M. (2021). Satellite mega-constellations create risks in low Earth orbit, the atmosphere and on Earth. *Scientific Reports*, 11(1), 10642. <https://doi.org/10.1038/s41598-021-89909-7>
- Caini, C., Cruickshank, H., Farrell, S., & Marchese, M. (2012). Delay- and disruption-tolerant networking (DTN): An alternative solution for future satellite networking applications. *Proceedings of the IEEE*, 99(11), 1980–1997. <https://doi.org/10.1109/JPROC.2011.2158378>
- Del Portillo, I., Cameron, B. G., & Crawley, E. F. (2019). A technical comparison of three LEO satellite constellation systems to provide global broadband connectivity. *Acta Astronautica*, 159, 216–225. <https://doi.org/10.1016/j.actaastro.2019.03.040>

DigitalGlobe. (2016). WorldView-3 data sheet. Maxar Technologies.

https://www.digitalglobe.com/assets/product-samples/DG2016_WorldView3_DataSheet.pdf

Elbert, B. R. (2008). Introduction to satellite communication (3rd ed.). Artech House.

ESA. (2019). European Data Relay System: Benefiting from optical satellite communications.

European Space Agency.

https://www.esa.int/Applications/Telecommunications_Integrated_Applications/EDRS

ESA. (2023). ESA's space environment report 2023. European Space Agency.

[https://www.esa.int/Space_Safety/Space_Debris/ESA_s_Space_Environment_Report_20](https://www.esa.int/Space_Safety/Space_Debris/ESA_s_Space_Environment_Report_2023)

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ETSI EN 302 307-2. (2015). Digital video broadcasting (DVB); second generation framing structure, channel coding and modulation systems for broadcasting, interactive services, news gathering and other broadband satellite applications; part 2: DVB-S2 extensions (DVB-S2X). European Telecommunications Standards Institute.

Evans, B. G., Kyrgiazos, A., & Thompson, P. (2005). Broadband satellite systems: Perspectives for the next decade. *International Journal of Satellite Communications and Networking*, 23(5), 283–300. <https://doi.org/10.1002/sat.824>

Falco, G., Guijt, W., & Parisi, G. (2018). Cybersecurity for the space industry: Threats, impacts, and responses. *IEEE Security & Privacy*, 16(5), 67–76. <https://doi.org/10.1109/MSP.2018.3761713>

Fenech, H., Amos, S., Tomatis, A., & Soumpholphakdy, V. (2016). High throughput satellite systems: An analytical approach. *IEEE Transactions on Aerospace and Electronic Systems*, 52(1), 192–202. <https://doi.org/10.1109/TAES.2015.140579>

Ferrus, R., Koumaras, H., Sallent, O., Agapiou, G., Rasheed, T., Kourtis, M. A., Boustie, C., Gelard, P., & Ahmed, T. (2016). SDN/NFV-enabled satellite communications networks: Opportunities, scenarios and challenges. *Physical Communication*, 18, 95–112. <https://doi.org/10.1016/j.phycom.2015.10.007>

Guerrero-Saade, J. A., & Guerrero-Saade, R. (2022). The Viasat incident: AcidRain wiper malware analysis. SentinelLabs. <https://www.sentinelone.com/labs/acidrain-a-modem-wiper-rains-down-on-europe/>

Humphreys, T. E., Ledvina, B. M., Psiaki, M. L., O'Hanlon, B. W., & Kintner, P. M. (2008).

Assessing the spoofing threat: Development of a portable GPS civilian spoofer.

Proceedings of the ION GNSS Conference, 2314–2325.

Ippolito, L. J. (2017). Satellite communications systems engineering: Atmospheric effects,

satellite link design and system performance (2nd ed.). Wiley.

ITU. (2023). Measuring digital development: Facts and figures 2023. International

Telecommunication Union. <https://www.itu.int/en/ITU->

[D/Statistics/Pages/facts/default.aspx](https://www.itu.int/en/ITU-D/Statistics/Pages/facts/default.aspx)

Kawamoto, Y., Nishiyama, H., Kato, N., Yoshimura, N., & Yamamoto, S. (2019). Toward future

unmanned aerial vehicle networks: Architecture, resource management, and experimental

fields. *IEEE Wireless Communications*, 26(1), 94–99.

<https://doi.org/10.1109/MWC.2018.1800121>

Kessler, D. J., & Cour-Palais, B. G. (1978). Collision frequency of artificial satellites: The

creation of a debris belt. *Journal of Geophysical Research*, 83(A6), 2637–2646.

<https://doi.org/10.1029/JA083iA06p02637>

Lin, X., Cioni, S., Charbit, G., Chuberre, N., Hellsten, S., & Boutillon, J. F. (2021). On the path to 6G: Embracing the next wave of low Earth orbit satellite access. *IEEE Communications Magazine*, 59(12), 36–42. <https://doi.org/10.1109/MCOM.001.2100298>

Liou, J. C., & Johnson, N. L. (2009). A sensitivity study of the effectiveness of active debris removal in LEO. *Acta Astronautica*, 64(2–3), 236–243. <https://doi.org/10.1016/j.actaastro.2008.07.009>

Maral, G., Bousquet, M., & Sun, Z. (2020). *Satellite communications systems: Systems, techniques and technology* (6th ed.). Wiley.

Morello, A., & Mignone, V. (2006). DVB-S2: The second generation standard for satellite broad-band services. *Proceedings of the IEEE*, 94(1), 210–227. <https://doi.org/10.1109/JPROC.2005.861013>

Moreira, A., Prats-Iraola, P., Younis, M., Krieger, G., Hajnsek, I., & Papathanassiou, K. P. (2013). A tutorial on synthetic aperture radar. *IEEE Geoscience and Remote Sensing Magazine*, 1(1), 6–43. <https://doi.org/10.1109/MGRS.2013.2248301>

Pratt, T., Bostian, C. W., & Allnutt, J. E. (2019). *Satellite communications* (3rd ed.). Wiley.

Roddy, D. (2006). Satellite communications (4th ed.). McGraw-Hill.

Roper, W. C., Kendall, F., & Walker, S. P. (2020). Hacking's a feature, not a bug: Lessons from the Blackjack programme for AI in satellite autonomy. Air Force Research Laboratory Technical Report AFRL-RV-PS-TR-2020-0158.

Saad, W., Bennis, M., & Chen, M. (2020). A vision of 6G wireless systems: Applications, enabling technologies, and design aspects. *IEEE Network*, 34(3), 134–142.
<https://doi.org/10.1109/MNET.001.1900287>

Satellite Industry Association. (2023). State of the satellite industry report 2023. SIA/Bryce Space and Technology. <https://www.sia.org/state-of-the-satellite-industry-report/>

Sklar, B. (2001). Digital communications: Fundamentals and applications (2nd ed.). Prentice Hall.

SpaceX. (2024). Starlink specifications and pricing. SpaceX.
<https://www.starlink.com/specifications>

Viasat Inc. (2023). ViaSat-3 satellite system overview. Viasat Technical Report.
<https://www.viasat.com/content/dam/viasat/news/pdfs/ViaSat-3-overview.pdf>

Vidal, O., Verelst, G., Lacan, J., Lagunas, E., Sala, J., & Caus, M. (2017). Next generation high throughput satellite system. IEEE First AESS European Conference on Satellite

Telecommunications (ESTEL), 1–7. <https://doi.org/10.1109/ESTEL.2012.6400139>

Wertz, J. R., & Larson, W. J. (2011). Space mission engineering: The new SMAD. Microcosm Press.